

AE-418: Radiation of [Ru(CO)(OH)₂(PNP)] in water with liquid phase O₂ measurment

Date: 2025-01-16

Tags: Radiation O₂ PNP

[Ru(CO)(OH)₂(PNP)] AE liquidphase

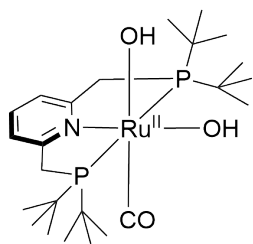
Category: Two-photon

Status: Done

Created by: Alexander Eith

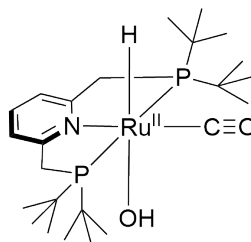
Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,64552
[Ru(CO)(OH)₂(PNP^tBu)]

365 nm (0.3 W)
white (0.2 W)
H₂O



Chemical Formula: C₂₄H₄₅NO₂P₂Ru
Molecular Weight: 542,64652
[Ru(CO)H(OH)(PNP^tBu)]

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Two-photon - AE-358: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH Two-photon - AE-386: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-382) 365 nm and white, 50 mM KOH in D ₂ O I

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	Two-photon - AE-382: Synthesis of [Ru(CO)(OH) ₂ (PNP ^t Bu)] using Ag ₂ O, 5 h (AE-382-3)	4.90	/	/	2.53	558.65	/
Milli-Q water	GD-018: Redegassing of H ₂ O	/	/	/	/	/	1

Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	10.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	1000
Irradiation start [s]	2775
Irradiation stop [s]	5330

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
15.01		A 10 mL screw cap schlenk flask was prepared	
	15:20	To the flask [Ru] was added under air	
		The flas was set under argon and stored in Equipment - Freezer Lab E004 CEEC II under Ar in the dark	
16.01		A screw cap 5 mL irradiation flask was prepared	
	9:57	water was added to the flask	

9:57 - 10:32	The suspension was stirred at 400 rpm	before filt.jpg brownish suspension
10:32	The suspension was filtered into the 5 mL flask using a PA syringe filter (pore diameter = 0.2 μm)	yellowish solution after filt.jpg
10:40	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power setup.jpg
	The Robust oxygen probe was attached to the flask using a bola fitting (diameter = 3 mm)	
11:05	The flask was moved to the setup	in setup.jpg
11:07	The logging was started (AE-418-1-Ch2)	
11:11	The flask was flushed with argon for 230 s (strong Ar flow in gas phase)	
11:14	The Schlenk was degassed according to Protocol - Degassing and O2 Measurement with a sensor cap with air-stable substances in the Photochem. setup	c(O2) = approx. -0.1 $\mu\text{mol/L}$ condensation in Schlenk flask.jpg small amount of condensed water was observed
11:49	The flask was flushed with argon for 4 min (strong Ar flow in gas phase)	c(O2) increased to approx. 0.5 $\mu\text{mol/L}$
11:54	The degassing stopped by removing the PTFE cannula according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	
12:00	The Log (-1-Ch2) was stopped	AE-418-1-Ch2.txt AE-418-1-Ch2.png
12:00	The logging was started (AE-418-2-Ch2)	
12:00 - 12:47	The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 2775 s
12:47	The light sources were turned on	irrad setup.jpg
13:29	The light source was turned off	Irradiation duration: 2555 s

	13:29 - 13:54	The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 1808 s
	13:54	The logging was stopped (-3-Ch2)	AE-418-2-Ch2.txt AE-418-2-Ch2.png after irradiation.jpg clear, orange solution
	14:54	The flask was opened to air and the pH value was measured using a pH paper (2.0 - 9.0)	AE-418-3 pH.jpg

Analysis

Date	Time	Sample name	Analysis method	Analytic device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
16.01	11:07	AE-418-1-Ch2	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-418-1-Ch2.txt	AE-418-1-Ch2.png	Two-photon - AE-402: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurement (AE-402-1-Ch2 and AE-402-2-Ch2)	Degassing did work in general. Fluctuations during degassing likely from Ar bubbles forming or disappearing at the tip of the robust probe. When degassing was switched to flushing: increase of O2 concentration from -0.15 µmol/L to 0.5 µmol/L. When flushing was stopped no change in rate of increase.
16.01	12:00	AE-418-2-Ch2	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-418-2-Ch2.txt	AE-418-2-Ch2.png	Two-photon - AE-402: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurement (AE-402-3-Ch2)	Very similar to AE-402. Slow decrease during equilibration. Sharp drop to 0.05 at start of irradiation, then gradual increase to 0.1 over approx. 5 min, then slow decrease similar to decrease before irradiation. At stop of irradiation drop to -0.05 µmol/L, then slow increase over time (maybe slowly flattening)
16.01	14:54	AE-418-3	pH Measurement	pH paper 2.0 - 9.0	H2O	pH.jpg	pH.jpg	/	pH = 5.5

Result

Linked experiments

- AE-236: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-237: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA I
- AE-245: Synthesis of poly(1,4-diphenylbutadiyne)
- AE-249: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-254: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA reproduction
- AE-257: Synthesis of poly(1,4-diphenylbutadiyne)
- AE-258: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-259: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 1 d
- AE-260: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 2 d
- AE-262: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 3 d
- AE-276: Synthesis of poly(1,4-diphenylbutadiyne), 1.2 g scale
- AE-279: Irradiation of PhPDA (AE-276-2), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-282: Irradiation of PhPDA (AE-276-1), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- GD-018: Redegassing of H₂O
- AE-313: Synthesis of PPy via Fenton reaction with FeSO₄ and 10 eq. H₂O₂ I
- AE-332: Irradiation of PPy (AE-312), 2 mg/mL
- AE-333: Irradiation of PPy (AE-313), 2 mg/mL
- C3N₄ - AE-398: Irradiation of C3N₄-Co-0.2 from Lingli Ni
- C3N₄ - AE-399: Irradiation of C3N₄-Co-0.4 from Lingli Ni I
- Two-photon - AE-358: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH
- Two-photon - AE-382: Synthesis of [Ru(CO)(OH)₂(PNPtBu)] using Ag₂O, 5 h

Two-photon - [AE-386: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-382\) 365 nm and white, 50 mM KOH in D2O I](#)

Two-photon - [AE-402: Radiation of \[Ru\(CO\)\(OH\)2\(PNP\)\] in water with liquid phase O2 measurment](#)

Linked resources

Archived - [Degassing procedure for photocatalytic water splitting by polymeric photocatalysts](#)

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Beam Combiner - [Beam Combiner LCS-BC25-0409 -2](#)

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 1\)](#)

Equipment - [Freezer Lab E004 CEEC II](#)

Light Source - [UHP LED 405 nm](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Light Source - [HP LED 405 -2](#)

Light Source - [HP LED 6500 K \(white\) -2](#)

Protocol - [Irradiation setup](#)

Protocol - [UV vis measurements \(BioTek PowerWave HT\)](#)

Protocol - [Determination of H2O2 by Spectroquant](#)

Protocol - [Degassing and O2 Measurement with sensor cap with air stable substances](#)

Protocol - [Determination of H2O2 in the suspensions of photocatalysts using H2O2 test strips](#)

Protocol - [Ultrasonication with the UP 200st from hilscher in ZAF 208](#)

Protocol - [H2O2 photometric detection with DPD-POD method for irradiated suspensions of photocatalysts](#)

Attached files

pH.jpg

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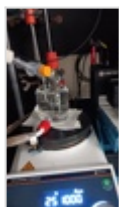
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in-setup.jpg

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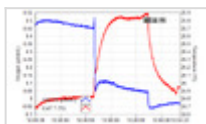


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AE-418-2-Ch2.png

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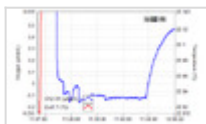


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